Evaporation

Pre-Lab Plan

Unit Operations Lab # 1

09/02/2025

Group # 2, Tuesday, AM Section

Team Members:

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1. Experimental Objective

The purpose of this experiment is to evaporate a sugar solution to analyze the capacity and economy of the evaporator used. This is done by using a vacuum pump connected to the loop containing the solution, which simultaneously controls the pressure, and hence the boiling point, in the separator. The water vapor is sent to a condenser to collect the water.

2. Experimental

This experiment was carried out using large-scale evaporation and fluid handling system to study how liquids can be heated and concentrated. This setup includes a main evaporation unit along with supporting equipment like pumps, flasks and gauges to monitor the process. The goal was to safely heat a feed solution, allow evaporation to occur and observe the system's behavior under controlled condition

a. Apparatus

Table 1. Apparatus for Evaporation Experiment

No.	Component	Description
1	Evaporation Chamber	Transparent chamber where the liquid is heated
2	Heater	Provides heat to the liquid to start the process
3	Pump	Helps in circulating the liquid through the system
4	Condenser	Carries vapor away for condensation
5	Gauges	Used to measure conditions inside the system
6	Storage containers	Used for extra feed or to collect the liquid

Figure 1: Single-effect Evaporator

b. Materials and Supplies:

Table 2. Materials and Supplies for Evaporation Experiment	
Materials & Supplies	**Description**
Sugar	Granulated White Sugar will be used to make a 10 wt% sugar water solution
Deionized Water	DI water will be fed to the apparatus in the makeup tank

	DI will be used to make enough Sugar Water solution to maintain a constant
Scale	Analytical balance (±0.05 g accuracy)
Funnels	For easy filling
Gloves and Lab Glasses	Safety
2000 mL Beaker	Beaker will be used to make and contain the Sugar Water solution

c. Experimental Plan

1. Ensure that the separator is fully empty by opening the valve at the bottom.
2. Prepare the sugar solution by adding 200 grams of sugar in 2000 mL of Deionized Water. Mix until the solution is homogeneous.
3. Fill the separator to the fill line using a funnel.
4. Fill the makeup tank with deionized water.
5. Start the flow of steam by twisting the back knob.
6. Start cooling water to the steam trap by twisting the bottom left valve.
7. Turn on the vacuum pump by switching on the device on the bottom. Adjust the valve to the desired pressure in the separator.
8. Wait until the temperature reaches a steady state, when the temperature increase reaches a plateau. Be sure to take measurements and regulate the volume of sugar water to the fill line throughout the process.
9. Once a steady state is reached, switch the outlet value from the now full Erlenmeyer flask to one that is empty.
10. Begin measuring the volume of the outlet water in respect to the time. Repeat after 5-10 minutes. Use the values to calculate the volume and time to calculate the flowrate.
11. Record the flow rate of the cooling water and the inlet and outlet temperature to determine the necessary energy inlet to the heat exchanger.
12. Repeat the process with a new vacuum level by changing the pressure. Three different pressure levels should be used overall.

Project Safety Evaluation

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: The wires can potentially be exposed to water/steam
Potential mechanical hazards?	No	If yes, identify conditions:
Potential pressure hazards?	Yes	If yes, identify conditions: Yes, working with a vacuum pump
Potential temperature hazards?	Yes	If yes, identify conditions: Steam will be at high temperature creating a possible temperature

		hazard, which is why the main apparatus remains behind a plexiglass barrier.
Hazardous/toxic chemicals involved?	No	If yes, identify conditions:
Gloves required?	Yes	If yes, identify conditions: Gloves will be required when handling and preparing sugar solution to avoid spilling on skin.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: White Granulated Sugar Steam
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	Not Specified	Heat Exchanger
Pressure (psia)	700 mmHg	Vacuum
Safety Controls	yes/no	If yes, Provide description

I have read relevant background material for the Unit Operations Laboratory entitled:
"____________________Evaporation__________________________" and understand the hazards associated

with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

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